Lab : 07

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Task 1:	

Code:	#include <iostream>
	#include <string>
	using namespace std;
	
	class Queue {
	public:
	string Arr[10];
	int Front;
	int rear;
	
	Queue() {
	Front = 0;
	rear = -1;
	}
	
	bool isEmpty() {
	if (rear < Front) {
	return true;
	} else {
	return false;
	}
	}
	
	bool isFull() {
	if (rear == 9) {
	return true;
	} else {
	return false;
	}
	}
	
	void Enqueue(string data) {
	if (isFull()) {
	cout << "Array is Overflow!" << endl;
	return;
	}
	rear++;
	Arr[rear] = data;
	}
	
	void Deque() {

```
        if (isEmpty()) {
            cout << "Array is Underflow " << endl;
            return;
        } else {
            cout << "Element '" << Arr[Front] << "' is deleted from front." << endl;
            Front++;
            return;
        }
    }

    void peek() {
        if (isEmpty()) {
            cout << "Array is Underflow/Empty " << endl;
            return;
        } else {
            cout << "First Element of array : " << Arr[Front] << endl;
            return;
        }
    }

    void Display() {
        if (isEmpty()) {
            cout << "Array is Underflow/Empty " << endl;
            return;
        } else {
            cout << "Elements of the array: " << endl;
            for (int i = Front; i <= rear; i++) {
                cout << "Element " << i + 1 << " : " << Arr[i] << endl;
            }
            return;
        }
    }
};

int main() {
    Queue q;
    int x;
    string inputData;

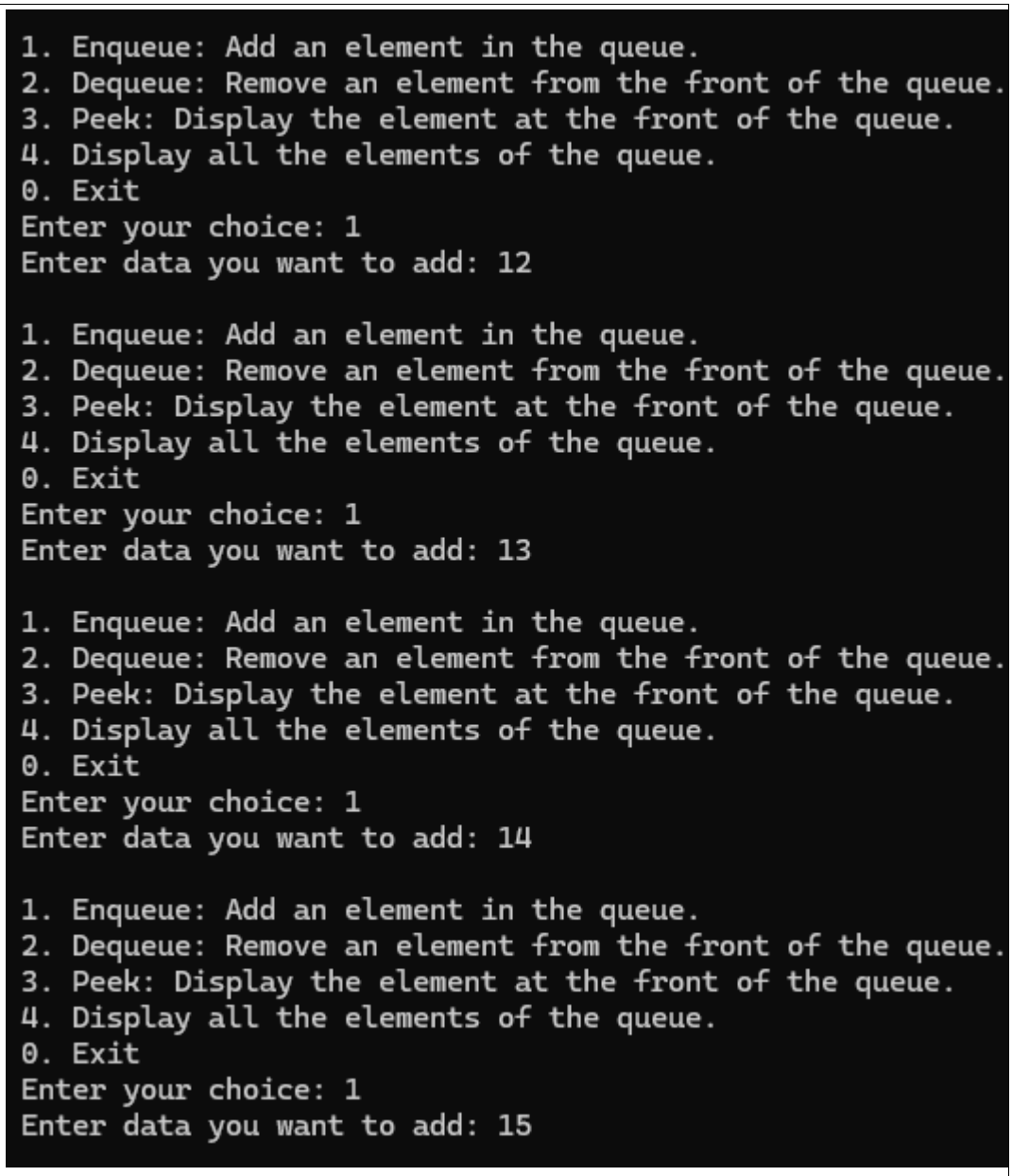
    do {
        cout << "\n1. Enqueue: Add an element in the queue. \n";
        cout << "2. Dequeue: Remove an element from the front of the queue.\n";
        cout << "3. Peek: Display the element at the front of the queue.\n";
        cout << "4. Display all the elements of the queue.\n";
        cout << "0. Exit " << endl;
        cout << "Enter your choice: ";
        cin >> x;

        if (x < 0 || x > 4) {
            cout << "Invalid input! TRY AGAIN " << endl;
```

```
        }

        if (x == 1) {
            if (q.isFull()) {
                cout << "Array is Overflow!" << endl;
            } else {
                cout << "Enter data you want to add: ";
                cin.ignore();
                getline(cin, inputData);
                q.Enqueue(inputData);
            }
        }
        if (x == 2) {
            q.Deque();
        }
        if (x == 3) {
            q.peek();
        }
        if (x == 4) {
            q.Display();
        }
    } while (x != 0);

    cout << "Program Ends! Thank you " << endl;
    return 0;
}
```

Screenshots:	

```
1. Enqueue: Add an element in the queue.
2. Dequeue: Remove an element from the front of the queue.
3. Peek: Display the element at the front of the queue.
4. Display all the elements of the queue.
0. Exit
Enter your choice: 2
Element '12' is deleted from front.

1. Enqueue: Add an element in the queue.
2. Dequeue: Remove an element from the front of the queue.
3. Peek: Display the element at the front of the queue.
4. Display all the elements of the queue.
0. Exit
Enter your choice: 3
First Element of array : 13

1. Enqueue: Add an element in the queue.
2. Dequeue: Remove an element from the front of the queue.
3. Peek: Display the element at the front of the queue.
4. Display all the elements of the queue.
0. Exit
Enter your choice: 4
Elements of the array:
Element 2 : 13
Element 3 : 14
Element 4 : 15

1. Enqueue: Add an element in the queue.
2. Dequeue: Remove an element from the front of the queue.
3. Peek: Display the element at the front of the queue.
4. Display all the elements of the queue.
0. Exit
Enter your choice: 0
Program Ends! Thank you
```

Task 2:	
Code:	#include <iostream> #include <string> using namespace std; class Queue { string* Arr; int capacity;

```
int Front;
int rear;
int Count;

public:
Queue(int s) {
   capacity = s;
   Arr = new string[capacity];
   Front = 0;
   rear = -1;
   Count = 0;
}

~Queue() {
   delete[] Arr;
}

bool isEmpty() {
   if (Count == 0) {
      return true;
   } else {
      return false;
   }
}

bool isFull() {
   if (Count == capacity) {
      return true;
   } else {
      return false;
   }
}

void Enqueue(string data) {
   if (isFull()) {
      cout << "Array is Overflow!" << endl;
      return;
   }
   rear = (rear + 1)%capacity;
   Arr[rear] = data;
   Count++;
}

void Deque() {
   if (isEmpty()) {
      cout << "Array is Underflow " << endl;
      return;
   } else {
      cout << "Element '" << Arr[Front] << "' is deleted from front." << endl;
      Front=(Front+1)%capacity;
```

```
            Count--;
            return;
        }
    }

    void peek() {
        if (isEmpty()) {
            cout << "Array is Underflow/Empty " << endl;
            return;
        } else {
            cout << "First Element of array : " << Arr[Front] << endl;
            return;
        }
    }

    void Display() {
        if (isEmpty()) {
            cout << "Array is Underflow/Empty " << endl;
            return;
        } else {
            int curr=Front;
            cout << "Elements of the array: " << endl;
            for (int i = 0; i < Count; i++) {
                cout << "Element " << i + 1 << " : " << Arr[curr] << endl;
                curr = (curr + 1)%capacity;
            }
            return;
        }
    }
};

int main() {

    int x;
    string inputData;
    cout<<"How long you want to make queue : ";
    cin>>x;
    Queue q(x);

    do {
        cout << "\n1. Enqueue: Add an element in the queue. \n";
        cout << "2. Dequeue: Remove an element from the front of the queue.\n";
        cout << "3. Peek: Display the element at the front of the queue.\n";
        cout << "4. Display all the elements of the queue.\n";
        cout << "0. Exit " << endl;
        cout << "Enter your choice: ";
        cin >> x;

        if (x < 0 || x > 4) {
            cout << "Invalid input! TRY AGAIN " << endl;
```

```
        }

        if (x == 1) {
            if (q.isFull()) {
                cout << "Array is Overflow!" << endl;
            } else {
                cout << "Enter data you want to add: ";
                cin.ignore();
                getline(cin, inputData);
                q.Enqueue(inputData);
            }
        }
        if (x == 2) {
            q.Deque();
        }
        if (x == 3) {
            q.peek();
        }
        if (x == 4) {
            q.Display();
        }
    } while (x != 0);

    cout << "Program Ends! Thank you " << endl;
    return 0;
}
```

Screenshots:

```
How long you want to make queue : 3

1. Enqueue: Add an element in the queue.
2. Dequeue: Remove an element from the front of the queue.
3. Peek: Display the element at the front of the queue.
4. Display all the elements of the queue.
0. Exit
Enter your choice: 1
Enter data you want to add: 1

1. Enqueue: Add an element in the queue.
2. Dequeue: Remove an element from the front of the queue.
3. Peek: Display the element at the front of the queue.
4. Display all the elements of the queue.
0. Exit
Enter your choice: 1
Enter data you want to add: 2

1. Enqueue: Add an element in the queue.
2. Dequeue: Remove an element from the front of the queue.
3. Peek: Display the element at the front of the queue.
4. Display all the elements of the queue.
0. Exit
Enter your choice: 1
Enter data you want to add: 3

1. Enqueue: Add an element in the queue.
2. Dequeue: Remove an element from the front of the queue.
3. Peek: Display the element at the front of the queue.
4. Display all the elements of the queue.
0. Exit
Enter your choice: 1
Array is Overflow!
```

```
1. Enqueue: Add an element in the queue.
2. Dequeue: Remove an element from the front of the queue.
3. Peek: Display the element at the front of the queue.
4. Display all the elements of the queue.
0. Exit
Enter your choice: 3
First Element of array : 1

1. Enqueue: Add an element in the queue.
2. Dequeue: Remove an element from the front of the queue.
3. Peek: Display the element at the front of the queue.
4. Display all the elements of the queue.
0. Exit
Enter your choice: 4
Elements of the array:
Element 1 : 1
Element 2 : 2
Element 3 : 3

1. Enqueue: Add an element in the queue.
2. Dequeue: Remove an element from the front of the queue.
3. Peek: Display the element at the front of the queue.
4. Display all the elements of the queue.
0. Exit
Enter your choice: 2
Element '1' is deleted from front.
```

```
1. Enqueue: Add an element in the queue.
2. Dequeue: Remove an element from the front of the queue.
3. Peek: Display the element at the front of the queue.
4. Display all the elements of the queue.
0. Exit
Enter your choice: 1
Enter data you want to add: 0

1. Enqueue: Add an element in the queue.
2. Dequeue: Remove an element from the front of the queue.
3. Peek: Display the element at the front of the queue.
4. Display all the elements of the queue.
0. Exit
Enter your choice: 4
Elements of the array:
Element 1 : 2
Element 2 : 3
Element 3 : 0

1. Enqueue: Add an element in the queue.
2. Dequeue: Remove an element from the front of the queue.
3. Peek: Display the element at the front of the queue.
4. Display all the elements of the queue.
0. Exit
Enter your choice: 0
Program Ends! Thank you
```

Task 3:

Code:

```
#include <iostream>
#include <string>
using namespace std;

class Queue {
private:
    string *Arr;
    int capacity;
    int n;
    int *priority;

public:
    Queue(int s) {
        capacity = s;
        Arr = new string[capacity];
        n = 0;
        priority = new int[capacity];
    }

    ~Queue() {
        delete[] Arr;
        delete[] priority;
    }
```

```
bool isEmpty() {
   if (n < 1) {
      return true;
   } else {
      return false;
   }
}

bool isFull() {
   if (n == capacity ) {
      return true;
   } else {
      return false;
   }
}

void Enqueue(string data, int prio) {
   if (isFull()) {
      cout << "Array is Overflow!" << endl;
      return;
   }
   int i = n - 1;
   while(i>=0 && priority[i]>prio)
   {
      Arr[i+1]=Arr[i];
      priority[i+1]=priority[i];
      i--;
   }
   Arr[i+1] = data;
   priority[i+1] = prio;
   n++;
   cout << data << " added with priority " << prio << endl;
}

void Deque() {
   if (isEmpty()) {
      cout << "Array is Underflow " << endl;
      return;
   } else {
     cout << Arr[0] << " (priority " << priority[0] << ") removed" << endl;
     for(int i=0; i<n-1; i++)
     {
       Arr[i]=Arr[i+1];
       priority[i]=priority[i+1];
     }
      n--;
      return;
   }
```

```
    }

    void Display() {
        if (isEmpty()) {
            cout << "Array is Underflow/Empty " << endl;
            return;
        } else {
            cout << "Elements of the array: " << endl;
            for (int i = 0; i < n; i++) {
                cout << "Element " << i + 1 << " Data : " << Arr[i] <<" , Priority :
"<<priority[i]<< endl;
            }
            return;
        }
    }
};

int main() {
    int x,pri;
    string inputData;
    cout<<"How long you want to make queue : ";
    cin>>x;
    Queue q(x);

    do {
        cout << "\n1. Enqueue: Add an element in the queue. \n";
        cout << "2. Dequeue: Remove an element from the front of the queue.\n";
        cout << "3. Display all the elements of the queue.\n";
        cout << "0. Exit " << endl;
        cout << "Enter your choice: ";
        cin >> x;

        if (x < 0 || x > 3) {
            cout << "Invalid input! TRY AGAIN " << endl;
        }

        if (x == 1) {
            if (q.isFull()) {
                cout << "Array is Overflow!" << endl;
            } else {
                cout << "Enter data you want to add: ";
                cin.ignore();
                getline(cin, inputData);
                cout << "Enter priority of your data: ";
                cin>>pri;
                q.Enqueue(inputData,pri);
            }
        }
        if (x == 2) {
            q.Deque();
```

```
        }

        if (x == 3) {
            q.Display();
        }
    } while (x != 0);

    cout << "Program Ends! Thank you " << endl;
    return 0;
}
```

Screenshots :

```
How long you want to make queue : 3

1. Enqueue: Add an element in the queue.
2. Dequeue: Remove an element from the front of the queue.
3. Display all the elements of the queue.
0. Exit
Enter your choice: 1
Enter data you want to add: Hania
Enter priority of your data: 2
Hania added with priority 2

1. Enqueue: Add an element in the queue.
2. Dequeue: Remove an element from the front of the queue.
3. Display all the elements of the queue.
0. Exit
Enter your choice: 1
Enter data you want to add: Sania
Enter priority of your data: 1
Sania added with priority 1

1. Enqueue: Add an element in the queue.
2. Dequeue: Remove an element from the front of the queue.
3. Display all the elements of the queue.
0. Exit
Enter your choice: 1
Enter data you want to add: Bushra
Enter priority of your data: 3
Bushra added with priority 3
```

```
1. Enqueue: Add an element in the queue.
2. Dequeue: Remove an element from the front of the queue.
3. Display all the elements of the queue.
0. Exit
Enter your choice: 3
Elements of the array:
Element 1 Data : Sania , Priority : 1
Element 2 Data : Hania , Priority : 2
Element 3 Data : Bushra , Priority : 3

1. Enqueue: Add an element in the queue.
2. Dequeue: Remove an element from the front of the queue.
3. Display all the elements of the queue.
0. Exit
Enter your choice: 2
Sania (priority 1) removed

1. Enqueue: Add an element in the queue.
2. Dequeue: Remove an element from the front of the queue.
3. Display all the elements of the queue.
0. Exit
Enter your choice: 1
Enter data you want to add: Jalal
Enter priority of your data: 1
Jalal added with priority 1
```

```
1. Enqueue: Add an element in the queue.
2. Dequeue: Remove an element from the front of the queue.
3. Display all the elements of the queue.
0. Exit
Enter your choice: 3
Elements of the array:
Element 1 Data : Jalal , Priority : 1
Element 2 Data : Hania , Priority : 2
Element 3 Data : Bushra , Priority : 3

1. Enqueue: Add an element in the queue.
2. Dequeue: Remove an element from the front of the queue.
3. Display all the elements of the queue.
0. Exit
Enter your choice: 0
Program Ends! Thank you
```